

Arrays, Multidimensional Matrices and Vectors in C++

> **Definition to Remember**

> Arra
autom

1. Concept Overview & Fundamentals

Arrays, Multidimensional Matrices and Vectors in C++ provides modern high-performance capabilities combining low-level hardware control with high-level abstractions, object-oriented paradigms, and Standard Template Library (STL) utilities.

In modern C++ (C++11 through C++20):

- **Z

2. Technical Architecture & Memory Mechanics

C++ Runtime Execution Architecture

```text

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### Modern C++ Feature Comparison

| Modern C++ Paradigm | Legacy Equivalent | Engineering Advantage | Performance Impact |

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3. Core Syntax & Implementation Patterns

Modern C++ Template Structure

```
```cpp
// Fast I/O and standard namespace configuration

std::cout << "Executing C++ Module: Arrays" << "\n";

return 0;
}
```

4. Real-World Applications & System Engineering

- **Quantitative Finance & HFT:** High-frequency trading platforms use modern C++ for microsecond-level execution.

5. Comprehensive Solved Problems & Code Walkthroughs

Problem 1: array max min

```
**Problem Description & Objective:**

**C++ Code Implementation:**
```cpp
// Implementation details...
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
#include
```

```
// Use standard library namespace globally
```

```
using n
```

```
// --- Main Execution Entry Point ---
```

```
int ma
```

```
// Evaluate conditional decision logic
```

```
if (n <=
```

```
// Declare dynamic C++ std::vector container
```

```
vector<
```

```
int minVal = arr[0];
```

```
int ma
```

```
// Loop execution block
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```

```

## Problem 2: array occurrences

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

```
/*
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
using namespace std;
```

```
// --- Main Execution Entry Point ---
```

```
int main()
```

```
// Output formatted data to console stream
```

```
cout << endl;
```

```
// Declare dynamic C++ std::vector container
```

```
vector<int> arr;
```

```
// Output formatted data to console stream
```

```
cout << endl;
```

```
// Loop execution block
```

```
for (int i = 0; i < arr.size(); i++)
```

```
// Output formatted data to console stream
```

```
cout << endl;
```

```
// Return zero code indicating successful program finish
```

```
return 0;
```

```

```

### **Problem 3: array palindrome**

**\*\*Problem Description & Objective:\*\***

Implement

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Declare dynamic C++ std::vector container
```

```
bool isPalin = true;
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```
return
```

```
---
```

Problem 4: array sorted check

****Problem Description & Objective:****

Implemen

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
#includ
```

```
// Use standard library namespace globally
```

```
using n
```

```
// --- Main Execution Entry Point ---
```

```
int mai
```

```
// Declare dynamic C++ std::vector container
```

```
vector<
```

```
bool isSorted = true;
```

```
// Loop
```

```
// Output formatted data to console stream
```

```
if (isSo
```

```
// Return zero code indicating successful program finish
```

```
return
```

```

```

### **Problem 5: array sum average**

**\*\*Problem Description & Objective:\*\***

Implem

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Declare dynamic C++ std::vector container
```

```
// Output formatted data to console stream
```

```
double average = sum / n;
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```
---
```

Problem 6: copy char array pointer

****Problem Description & Objective:****

****C++ Code Implementation:****

/*

#include <iostream>

// Use standard library namespace globally

void copyStringPointer(const char* src, char* dest) {

// --- Main Execution Entry Point ---

copyStringPointer(source, destination);

// Output formatted data to console stream

```
// Return zero code indicating successful program finish
```

```
return
```

```
---
```

Problem 7: delete array element

****Problem Description & Objective:****

Implemen

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
#includ
```

```
// Use standard library namespace globally
```

```
using n
```

```
// --- Main Execution Entry Point ---
```

```
int mai
```

```
// Output formatted data to console stream
```

```
cout <
```

```
// Declare dynamic C++ std::vector container
```

```
// Output formatted data to console stream
```

```
// Declare dynamic C++ std::vector container
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```

```

### **Problem 8: diagonal sum**

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

```
/*
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Declare dynamic C++ std::vector container
```

```
// Output formatted data to console stream
```

```
int primaryDiagonal = 0;
```

```
// Loop execution block
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```
return
```

```

```

### Problem 9: merge sorted arrays

**\*\*Problem Description & Objective:\*\***

Implemen

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
#includ
```

```
// Use standard library namespace globally
```

```
using n
```

```
// --- Main Execution Entry Point ---
```

```
int mai
```

```
// Output formatted data to console stream
```

```
cout <
```

```
// Output formatted data to console stream
```

```
cout <
```

```
// Declare dynamic C++ std::vector container
```

```
vector<
```

```
// Loop execution block
```

```
while (
```

```
// Loop execution block
```

```
while (
```

```
// Output formatted data to console stream
```

```
cout <
```

```
// Return zero code indicating successful program finish
```

```
return
```

```
---
```

Problem 10: reverse array

****Problem Description & Objective:****

Implem

****C++ Code Implementation:****

```cpp

/\*

- ===

#include <iostream>

#includ

// Use standard library namespace globally

using n

// --- Main Execution Entry Point ---

int mai

// Declare dynamic C++ std::vector container

vector<

int start = 0, end = n - 1;

```
// Loop
```

```
// Output formatted data to console stream
```

```
cout <<
```

```
// Return zero code indicating successful program finish
```

return

...

### Problem 11: search 2d array

**\*\*Problem Description & Objective:\*\***

## Implement

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

$$/^{*}$$

— — —

```
#include <iostream>
```

```
#include
```

```
// Use standard library namespace globally
```

```
using n
```

```
// --- Main Execution Entry Point ---
```

```
int ma
```

```
// Output formatted data to console stream
```

```
cout <
```

```
// Declare dynamic C++ std::vector container
```

```
vector<
```

```
// Output formatted data to console stream
```

```
cout <
```

```
// Output formatted data to console stream
```

```
cout <
```

```
bool found = false;
```

```
// Loop
```

```
// Output formatted data to console stream
```

```
if (!fou
```

```
// Return zero code indicating successful program finish
```

return

Problem 12: sum average 2d array****Problem Description & Objective:****

Implem

****C++ Code Implementation:****

```cpp

/\*

- ===

#include &lt;iostream&gt;

#includ

// Use standard library namespace globally

using n

// --- Main Execution Entry Point ---

int mai

// Output formatted data to console stream

cout &lt;

// Declare dynamic C++ std::vector container

```
// Output formatted data to console stream
```

```
int totalElements = rows * cols;
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```

```

## 6. Exam Scoring Points & Memory Revision

```
> Must-Write Points
```

```
> Quick Recall
```